CONFIDENTIAL – EXAMINER'S MODEL ANSWER KEY & MARKING SCHEME

SUBJECT: DATABASE MANAGEMENT SYSTEMS & INFORMATION ARCHITECTURE | CODE: CS-402
TOTAL MARKS: 100 | EVALUATION RUBRIC

Q.1: Demonstrate the use of GROUP BY and HAVING clauses in SQL. Differentiate between WHERE and HAVING with queries. [20 Marks]

Evaluation Component	Expected Details & Examiner Criteria	Marks
Formal Definition & Theoretical Foundation	Accurate formal definition, taxonomy, and mathematical/formal properties.	**4M**
System Architecture / Structural Diagram	Neat, properly labeled architectural diagram or ER/schema representation.	**5M**
Detailed Technical Explanation & Working Mechanism	Comprehensive explanation of operational algorithm, constraints, or protocols.	**7M**
Practical Case Study / SQL Queries / Numerical Demonstration	Realistic implementation example, code snippet, or anomaly walk-through.	**4M**

Deductions / Common Mistakes: Giving vague generic answers without subject-specific technical terminology (-2 Marks) | Omitting the diagram or failing to label components clearly (-3 Marks) | Failing to provide a working code or concrete anomaly example (-2 Marks)

Q.2: Discuss ACID properties in transaction processing. Explain how each property is preserved by the DBMS recovery manager. [20 Marks]

Evaluation Component	Expected Details & Examiner Criteria	Marks
Formal Definition & Theoretical Foundation	Accurate formal definition, taxonomy, and mathematical/formal properties.	**4M**
System Architecture / Structural Diagram	Neat, properly labeled architectural diagram or ER/schema representation.	**5M**
Detailed Technical Explanation & Working Mechanism	Comprehensive explanation of operational algorithm, constraints, or protocols.	**7M**
Practical Case Study / SQL Queries / Numerical Demonstration	Realistic implementation example, code snippet, or anomaly walk-through.	**4M**

Deductions / Common Mistakes: Giving vague generic answers without subject-specific technical terminology (-2 Marks) | Omitting the diagram or failing to label components clearly (-3 Marks) | Failing to provide a working code or concrete anomaly example (-2 Marks)

Q.3: Explain the Strict Two-Phase Locking (2PL) protocol. How does 2PL ensure serializability? [20 Marks]

Evaluation Component	Expected Details & Examiner Criteria	Marks
Formal Definition & Theoretical Foundation	Accurate formal definition, taxonomy, and mathematical/formal properties.	**4M**
System Architecture / Structural Diagram	Neat, properly labeled architectural diagram or ER/schema representation.	**5M**
Detailed Technical Explanation & Working Mechanism	Comprehensive explanation of operational algorithm, constraints, or protocols.	**7M**
Practical Case Study / SQL Queries / Numerical Demonstration	Realistic implementation example, code snippet, or anomaly walk-through.	**4M**

Deductions / Common Mistakes: Giving vague generic answers without subject-specific technical terminology (-2 Marks) | Omitting the diagram or failing to label components clearly (-3 Marks) | Failing to provide a working code or concrete anomaly example (-2 Marks)

Q.4: Describe Deadlock handling mechanisms: Wait-Die scheme, Wound-Wait scheme, and Wait-For Graph (WFG) detection. [20 Marks]

Evaluation Component	Expected Details & Examiner Criteria	Marks

Formal Definition & Theoretical Foundation	Accurate formal definition, taxonomy, and mathematical/formal properties.	**4M**
System Architecture / Structural Diagram	Neat, properly labeled architectural diagram or ER/schema representation.	**5M**
Detailed Technical Explanation & Working Mechanism	Comprehensive explanation of operational algorithm, constraints, or protocols.	**7M**
Practical Case Study / SQL Queries / Numerical Demonstration	Realistic implementation example, code snippet, or anomaly walk-through.	**4M**

Deductions / Common Mistakes: Giving vague generic answers without subject-specific technical terminology (-2 Marks) | Omitting the diagram or failing to label components clearly (-3 Marks) | Failing to provide a working code or concrete anomaly example (-2 Marks)

Q.5: Discuss mapping cardinalities (1:1, 1:N, M:N) and structural constraints in ER diagrams. [20 Marks]

Evaluation Component	Expected Details & Examiner Criteria	Marks
Formal Definition & Theoretical Foundation	Accurate formal definition, taxonomy, and mathematical/formal properties.	**4M**
System Architecture / Structural Diagram	Neat, properly labeled architectural diagram or ER/schema representation.	**5M**
Detailed Technical Explanation & Working Mechanism	Comprehensive explanation of operational algorithm, constraints, or protocols.	**7M**
Practical Case Study / SQL Queries / Numerical Demonstration	Realistic implementation example, code snippet, or anomaly walk-through.	**4M**

Deductions / Common Mistakes: Giving vague generic answers without subject-specific technical terminology (-2 Marks) | Omitting the diagram or failing to label components clearly (-3 Marks) | Failing to provide a working code or concrete anomaly example (-2 Marks)